Chunhui Zhang, CS Ph.D. candidate at Dartmouth

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Research Focus

RL Multimodal LLMs on Language, vision, and audio

Education

Dartmouth College Hanover, NH, USA
Ph.D. candidate in Computer Science Aug. 2023–Jun. 2026 (expected)
Advisor: Professor Soroush Vosoughi

Brandeis University Waltham, MA, USA
M.S. in Computer Science Sep. 2021–Jun. 2023
GSAS Research Fellowship

Northeastern University China
B.S. in Computer Science Sep. 2017–Jun. 2021
Outstanding Honor Thesis Award

Experience

Amazon Seattle, WA, USA
Applied Scientist Intern Sep. 2025 – Dec. 2025
RL Post-Training for Multimodal Computer-Use Agents (CUA) in Rufus
Post-trained multimodal LLMs (7–32B) for low-latency GUI agents towards **adaptive reasoning** with reduced token costs.
Developed verl and vLLM to enable **on-policy RL** and **SFT** on complex trajectories, bridging OOD gaps and stabilizing visual grounding.

Google DeepMind Mountain View, CA, USA
Research Intern Jun. 2025 – Sep. 2025
Gemma 3n Post-Training for Real-Time Streaming (Frontier AI & GenAI)
Created a synthetic audio–text framework for real-time ASR/AST (speech/voice tasks); eliminated human labeling dependence while keeping WER/BLEU within **1%** of baselines.
Built the **full-stack RL+SFT infrastructure** for Gemma 3n, enabling PyTorch/SGLang support for the open-source community.
Engineered a high-throughput training pipeline for Gemma 3n, accelerating RL (GRPO, DAPO) by 5×.
Contributed to the **Gemma 3n technical report** and open-source development.

Honda Research Institute USA San Jose, CA, USA

Research Intern Jun. 2024–Sep. 2024

Multimodal LLM Post-Training (8B–70B) for Social Reasoning

Published a *spotlight* paper at ICML 2025 (Top 2.59%). Filed a US patent. Developed long-context multimodal LLMs and accelerated inference-time scaling using vLLM framework. Enabled long-context modeling of human behavior and autonomous driving in multimodal environments; scalable to LLMs up to 70B/405B parameters.

Selected Publications

Overcoming Multi-step Complexity in Theory-of-Mind Reasoning: A Scalable Bayesian Planner

Chunhui Zhang, Zhongyu Ouyang, Sean Dae Houlihan, Kwonjoon Lee, Nakul Agarwal, Soroush Vosoughi, Shao-Yuan Lo

ICML 2025 ***(Spotlight, Top 2.59%)***

Paper | Code

First scalable solution for multi-step Theory-of-Mind reasoning. Uses Bayesian inverse planning for global planning, then lets LLMs focus on local reasoning. Works on 70B+ models where others fail.

Growing Through Experience: Scaling Episodic Grounding in Language Models

Chunhui Zhang, Elsie Wang, Zhongyu Ouyang, Xiangchi Yuan, Soroush Vosoughi

ACL 2025 Main Conference ***(Oral Presentation, Top 3.24%)***

Paper | Code

Post-trained reasoning LLMs (across 3B, 8B, 70B) on MCTS-sampled data from physical simulators.

Pretrained Image-Text Models are Secretly Video Captioners

Chunhui Zhang*, Yiren Jian*, Zhongyu Ouyang, Soroush Vosoughi

NAACL 2025 Main Conference ***(Oral Presentation, Top 2.88%)***

Paper | Code

RL post-training recipe that achieved **Top-2** on PapersWithCode video captioning leaderboard, outperforming industry MLLM captioners.

Knowing More, Acting Better: Hierarchical Representation for Embodied Decision-Making

Chunhui Zhang, Zhongyu Ouyang, Zheyuan Liu, Soroush Vosoughi

EMNLP 2025 Findings

Paper

Refines vision-language-action LLM representations to enable more effective PPO-based RL training in embodied AI.

Working Memory Identifies Reasoning Limits in Language Models
Chunhui Zhang, Yiren Jian, Zhongyu Ouyang, Soroush Vosoughi
EMNLP 2024 Main Conference | Paper | Code
Introduced working memory as a diagnostic tool for LLM reasoning limits. This work inspired a follow-up NAACL paper on long-context multimodal understanding.

Temporal Working Memory: Query-Guided Segment Refinement for Enhanced Multimodal Understanding
Xingjian Diao*, **Chunhui Zhang***, Weiyi Wu, Zhongyu Ouyang, Peijun Qing, Ming Cheng, Soroush Vosoughi, Jiang Gui
NAACL 2025 Findings | Paper | Code
Inspired by working memory in my EMNLP 2024 paper, this work is a follow-up study on long-context video-language understanding.

Under Review

Model Priors Shape Experience: RL for Complex Audio Long-form Reasoners
Chunhui Zhang
Code
First to implement Qwen2.5-Omni and vLLM for faster RL reasoning across audio and other unified modalities.

Selected Collaborations

Superficial Self-Improved Reasoners Benefit from Model Merging
Xiangchi Yuan, **Chunhui Zhang**, Zheyuan Liu, Dachuan Shi, Soroush Vosoughi, Wenke Lee
EMNLP 2025 | Paper | Code

SoundMind: RL-Incentivized Logic Reasoning for Audio-Language Models
Xingjian Diao, **Chunhui Zhang**, Keyi Kong, Weiyi Wu, Chiyu Ma, Zhongyu Ouyang, Peijun Qing, Soroush Vosoughi, Jiang Gui
EMNLP 2025 ***(Oral Presentation)***

Is It Navajo? Accurate Language Detection in Endangered Athabaskan Languages
Ivory Yang, Weicheng Ma, **Chunhui Zhang**, Soroush Vosoughi
NAACL 2025 ***(Oral Presentation, Top 2.88%)*** | Paper | Code

Learning Musical Representations for Music Performance Question Answering
Xingjian Diao, **Chunhui Zhang**, Tingxuan Wu, Ming Cheng, Zhongyu Ouyang, Weiyi Wu, Jiang Gui
EMNLP findings 2024

Expedited Training of Visual Conditioned Language Generation via Redundancy Reduction
Yiren Jian, Tingkai Liu, Yunzhe Tao, **Chunhui Zhang**, Soroush Vosoughi, Hongxia Yang
ACL 2024 ***(Oral Presentation, Top 3.10%)***

Aligning Relational Learning with Lipschitz Fairness
Yaning Jia*, **Chunhui Zhang***, Soroush Vosoughi
ICLR 2024 – Note: Co-first author Jia was a master student I mentored.

Mitigating Emergent Robustness Degradation on Graphs while Scaling-up
Xiangchi Yuan*, **Chunhui Zhang***, Yijun Tian, Yanfang Ye, et al.
ICLR 2024 – Note: Co-first author Yuan was a master student I mentored.

When Sparsity Meets Contrastive Models: Less Data Can Bring Better Class-Balanced Representations
Chunhui Zhang, Chao Huang, Yijun Tian, Qianlong Wen, et al.
ICML 2023 – **AAAI-DCAA 2023 Best Paper Runner-up Award**

Chasing All-Round Graph Representation Robustness: Model, Training, and Optimization
Chunhui Zhang, Yijun Tian, Mingxuan Ju, Zheyuan Liu, et al.
ICLR 2023

Look Twice as Much as You Say: Scene Graph Contrastive Learning for Self-Supervised Image Caption Generation
Chunhui Zhang, Chao Huang, Youhuan Li, Xiangliang Zhang, Yanfang Ye, et al.
CIKM, 2022.

Full publication list *includeing 30+ papers at ICML, NeurIPS, ICLR, ACL, NAACL, EMNLP, WWW (TheWebConf), KDD, CIKM, ICDM, ICWSM, etc. available at chunhuizng.github.io*

Honors and Awards	ICML Spotlight Recognition (Top 2.59%)	2025
	ACL Oral Presentation Award (Top 3.24%)	2024, 2025
	NAACL Oral Presentation Award (Top 2.88%)	2025
	AAAI-DCAA Best Paper Runner-up Award	2023
	Graduate School of Arts and Sciences Fellowship	2021–2023

Service

Program Committee/Reviewer: ICML, ICLR, NeurIPS, ACL, NAACL, EMNLP, UAI, AISTATS, AAAI (2023–2025)

Journal Reviewer: TMLR, IEEE TKDE, IEEE TNSE, ACM TIST, Neurocomputing